Introduction to Econometrics
Week3: Hypothesis testing part 2
(Hayashi, chapter 1.4)

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Based on Rob Alessie lecture notes

Organization of the lecture

1. Continuation of Hypothesis Testing under Normality
   - Linear Hypotheses
   - The F-Test
   - A more convenient expression for the F test
   - t versus F
2. Chow test

Linear hypotheses

- The null hypothesis is often about linear combinations of restrictions written as a system of linear equations:

$$\mathbf{R}\beta = \mathbf{r} \tag{1}$$

  where $\mathbf{R}$ and $\mathbf{r}$ are known and specified by the hypothesis.
- $\mathbf{R}$ is a $\#r \times K$ matrix where $\#r$ denotes the number of restrictions on the regression coefficients.
- $\mathbf{R}$ specifies where the restriction is placed
- $\mathbf{r}$ is a $(\#r \times 1)$-vector that specifies the restriction
- $H_1 : H_0$ not true
- Obviously, matrix $\mathbf{R}$ should have full row rank,i.e $rank(\mathbf{R}) = \#r$. In other words one should not include redundant equations (restrictions).
- Obviously, $\#r \leq K$

Example revisited (wage equation)

- We'd like to check whether years of experience has an effect on the hourly wage in the following model:

$$\log(wage_i) = \beta_1 + \beta_2 educ_i + \beta_3 exper_i + \beta_4 exper_i^2 + \varepsilon_i \qquad (2)$$

- So

$$H_0 : \beta_3 = 0, \beta_4 = 0$$

- $$H_1 : \beta_3 \neq 0 \text{ or } \beta_4 \neq 0$$

- In this case,

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Notice that the matrix $\mathbf{R}$ has full row rank.

Example revisited (redundant restrictions)

- One could also formulate H_0 as:

$$H_0 : \beta_3 = 0, \beta_4 = 0, \beta_3 = \beta_4$$

- Notice that one of these restrictions is redundant
- In this case the matrices $\mathbf{R}$ and $\mathbf{r}$ are equal to

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- Here $rank(\mathbf{R}) \neq 3$ because the third can be written as a linear combination of the first and second row.
- Obviously one can also add a restriction on the *educ* coefficient, e.g. $\beta_2 = 0.05$ In this case $\mathbf{R}$ and $\mathbf{r}$ become

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ 0.05 \end{bmatrix}$$

The F-test

- **Proposition 1.4 (distribution of the F-ratio):** *Suppose Assumptions 1.1-1.5 hold. Under the null hypothesis* $H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ where $\mathbf{R}$ is $\#r \times K$ with $rank(\mathbf{R}) = \#r$, the F-ratio defined as:

$$\begin{aligned} F &\equiv \frac{(\mathbf{Rb} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{Rb} - \mathbf{r})/\#\mathrm{r}}{s^2} \\ &= (\mathbf{Rb} - \mathbf{r})'[\mathbf{R}\widehat{var}(\mathbf{b}|\mathbf{X})\mathbf{R}']^{-1}(\mathbf{Rb} - \mathbf{r})/\#\mathrm{r} \end{aligned} \tag{3}$$

is distributed as $F(\#r, n - K)$ (the F distribution with $\#r$ and $n - K$ degrees of freedom).

The F-test

Proof proposition 1.4

- As in Proposition 1.3, it suffices to show that

$$F|\mathbf{X} \sim F(\#r, n-K)$$

  because the F distribution does not depend on $\mathbf{X}$, it is also the unconditional distribution of the statistic.
- *Fact 4* Let $\mathbf{x}$ be an m dimensional random vector. If $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ a nonsingular, then

$$(\mathbf{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \sim \chi^2(m)$$

The F-test

Proof proposition 1.4

- Since $s^2 = \mathbf{e}'\mathbf{e}/(n-K)$, we can write F as

$$F = \frac{w/\#r}{q/(n-K)}$$

where $q = \mathbf{e}'\mathbf{e}/\sigma^2$ and

$$w = (\mathbf{Rb} - \mathbf{r})'[\sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{Rb} - \mathbf{r})$$

We need to show

1) $w|\mathbf{X} \sim \chi^2(\#r)$
2) $q|\mathbf{X} \sim \chi^2(n-K)$ (this is part 1 in the proof of theorem 1.3)
3) w and q are independently distributed conditional on $\mathbf{X}$.

Then, by the definition of the F distribution,

$$F \sim F(\#r, n-K)$$

The F-test

Proof proposition 1.4

- Let $\mathbf{v} = \mathbf{Rb} - \mathbf{r}$. Under H_0: $\mathbf{Rb} - \mathbf{r} = \mathbf{R}(\mathbf{b} - \beta)$
  We already know that $\mathbf{b} - \beta|\mathbf{X} \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ So conditional on $\mathbf{X}$, $\mathbf{v}$ is normal with mean $\mathbf{0}$, and variance

$$\begin{aligned} var(\mathbf{v}|\mathbf{X}) &= var(\mathbf{R}(\mathbf{b} - \beta)|\mathbf{X}) = \mathbf{R}var(\mathbf{b} - \beta|\mathbf{X})\mathbf{R}' = \\ &= \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}' \end{aligned}$$

  which is the inverse of the middle matrix in the quadratic form for w. Hence,

$$w = \mathbf{v}'var(\mathbf{v}|\mathbf{X})^{-1}\mathbf{v}$$

The F-test

Proof proposition 1.4

- Since $\mathbf{R}$ is of full row rank and $\mathbf{X}'\mathbf{X}$ is nonsingular (assumption 1.3), then $Var(\mathbf{v}|\mathbf{X}) = \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is nonsingular (and we can therefore take an inverse in equation above)
- Moreover due to fact 4, $w|\mathbf{X} \sim \chi^2(\#r)$
- w is a function of $\mathbf{b}$ and q is a function of $\mathbf{e}$.
- But $\mathbf{b}$ and $\mathbf{e}$ are independently distributed conditional on $\mathbf{X}$ (see proof proposition 1.3)

So w and q are independently distributed conditional on $\mathbf{X}$.
q.e.d.

The F-test: intuition

$$F = (\mathbf{Rb} - \mathbf{r})'[\mathbf{R}\widehat{var}(\mathbf{b}|\mathbf{X})\mathbf{R}']^{-1}(\mathbf{Rb} - \mathbf{r})/\#\mathrm{r}$$

- If $H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is true, we expect $\mathbf{Rb} - \mathbf{r}$ to be small
- So large F should be taken as evidence for a failure of the null.
- Look therefore at right tail of the F-distribution, see figure 1.4 of the book

The F-test

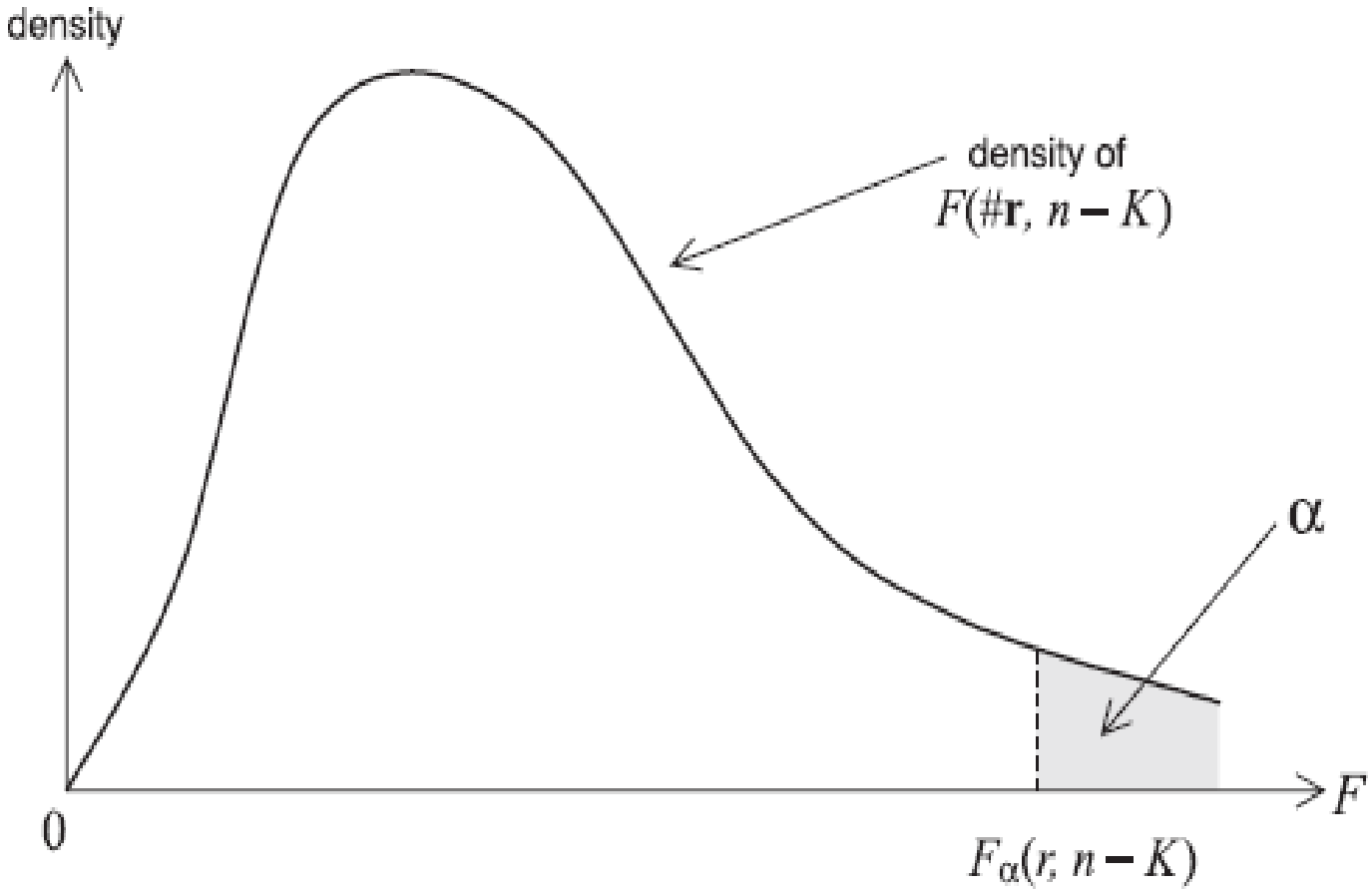

Figure 1.4: F Distribution

The F-test

Steps F test when using critical value:

1. Formulate H_0 and H_1 (see above).
2. Specify α.
3. Calculate the F-ratio.
4. Go to the table of F distribution and look up the entry for **#r** (the numerator degrees of freedom) and $n - K$ (the denominator degrees of freedom). Find the critical value $F_\alpha(\#r, n - K)$ as illustrated in Figure 1.4. For example, when $\#r = 3$, $n - K = 30$ and $\alpha = 0.05$, then $F_{0.05}(3, 30) = 2.92$.
5. Do not reject H_0 if $F < F_\alpha(\#r, n - K)$, otherwise reject H_0.

The F-test

Steps F test when using p-Values:

1. same
2. same
3. same
4. Calculate
   p=area of the upper tail of the F distribution to the right of the F-ratio:
5. Do not reject H_0 if $p > \alpha$; reject otherwise.

A More Convenient Expression for F

- The above derivation of the F-ratio is by the **Wald principle**: it is based on the unrestricted estimator (see equation (3)).
- A more convenient formula is available, involving two different sum of squared residuals:
  1. The 'unrestricted' sum of squares' $SSR_U(\mathbf{b}) = \mathbf{e}'\mathbf{e}$ based on the unrestricted regression $\mathbf{b}$= OLS estimator of the unrestricted model.
  2. the 'restricted' sum of squares' $SSR_R(\hat{\boldsymbol{\beta}})$ obtained from

$$\min_{\tilde{\boldsymbol{\beta}}} SSR(\tilde{\boldsymbol{\beta}}) \text{ s.t. } \mathbf{R}\tilde{\boldsymbol{\beta}} = \mathbf{r} \tag{4}$$

  ($\hat{\boldsymbol{\beta}}$ is the solution of this minimization problem).
- Finding the $\tilde{\boldsymbol{\beta}}$ that achieves this constrained minimization is called the **restricted regression** or **restricted least squares**.
- **Intuition**: if restrictions are true, the increase in sum of squares should not be too large relative to the influence of the error

A More Convenient Expression for F

- **Proposition 1.4.a**: The F-ratio (3) can be rewritten as

$$F = \frac{(SSR_R - SSR_U)/\#r}{SSR_U/(n-K)} = \frac{(SSR_R - SSR_U)/\#r}{s^2} \tag{5}$$

- This second derivation of the F-ratio is said to be by the **Likelihood-Ratio principle**. (one needs to estimate both the restricted and the unrestricted model).
- Equation (5) can be rewritten as follows:

$$F = \frac{(R_U^2 - R_R^2)/\#r}{(1 - R_U^2)/(n-K)} \tag{6}$$

A More Convenient Expression for F

Proof proposition 1.4.a (in two steps)

Step 1: Derive the restricted least squares estimator
Lagrangian of minimization problem (4):

$$\begin{aligned} L(\tilde{\beta}) &= 0.5(\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta}) - \lambda'(\mathbf{r} - \mathbf{R}\tilde{\beta}) \\ &= 0.5(\mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta}) - \lambda'(\mathbf{r} - \mathbf{R}\tilde{\beta}) \end{aligned}$$

Derive the first order conditions (check!)

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} - \mathbf{R}'\lambda \Leftrightarrow \hat{\beta} = \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\lambda \tag{7}$$

$$\mathbf{R}\hat{\beta} = \mathbf{r} \tag{8}$$

A More Convenient Expression for F

Pre-multiplying equation (7) with $\mathbf{R}$ and using equation (8) gives

$$\begin{gathered}\mathbf{Rb} - \mathbf{r} = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\boldsymbol{\lambda} \;\Leftrightarrow \\ \boldsymbol{\lambda} = (\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')^{-1}(\mathbf{Rb} - \mathbf{r})\end{gathered}$$

Substituting this into equation (7)

$$\mathbf{b} - \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'(\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')^{-1}(\mathbf{Rb} - \mathbf{r}) \tag{9}$$

A More Convenient Expression for F

Step2: Derivation of formula (5) (given (3))

- Let $\hat{\varepsilon} = \mathbf{y} - \mathbf{X}\hat{\beta}$, the residuals of the restricted regression.
- Obviously $\hat{\varepsilon} = \mathbf{y} - \mathbf{X}\mathbf{b} + \mathbf{X}(\mathbf{b} - \hat{\beta}) = \mathbf{e} + \mathbf{X}(\mathbf{b} - \hat{\beta})$

Then

$$\begin{aligned}
SSR_R &= \hat{\varepsilon}'\hat{\varepsilon} = (\mathbf{e} + \mathbf{X}(\mathbf{b} - \hat{\beta}))'(\mathbf{e} + \mathbf{X}(\mathbf{b} - \hat{\beta})) \\
&= SSR_U + 2(\mathbf{b} - \hat{\beta})'\mathbf{X}'\mathbf{e} + (\mathbf{b} - \hat{\beta})'(\mathbf{X}'\mathbf{X})(\mathbf{b} - \hat{\beta}) \Leftrightarrow \\
SSR_R - SSR_U &= (\mathbf{b} - \hat{\beta})'(\mathbf{X}'\mathbf{X})(\mathbf{b} - \hat{\beta}) \Leftrightarrow \\
SSR_R - SSR_U &= (\mathbf{R}\mathbf{b} - \mathbf{r})'(\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')^{-1}(\mathbf{R}\mathbf{b} - \mathbf{r}) \qquad (10)
\end{aligned}$$

In the last step I use equation (9) (I leave this check as an exercise)
Substituting (10) into (3) gives (notice that
$s^2 = SSR_U/(n-K)$):

$$F \equiv \frac{(SSR_R - SSR_U)/\#\mathrm{r}}{SSR_U/(n-K)}$$

q.e.d.

t versus F

- Because hypotheses about individual coefficients are linear hypotheses, the t-test of $H_0 : \beta_k = \bar{\beta}_k$ is a special case of the F-test.
- To see this, note that the hypothesis can be written as $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with

$$\mathbf{R} = [0 \ldots, 0, 1, 0 \ldots 0] \ , \ \mathbf{r} = \bar{\beta}_k$$

- So by (3) the F-ratio is

$$(b_k - \bar{\beta}_k)'[s^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}]^{-1}(b_k - \bar{\beta}_k)$$

which is the square of the t-ratio. So $F(1, n-K) = t^2_{n-K}$.

t versus F

One should not drop variables from the model based on their individual insignificance: it is possible that two variables are individually insignificant (cf. t-values), but they are jointly significant(cf. F-test).

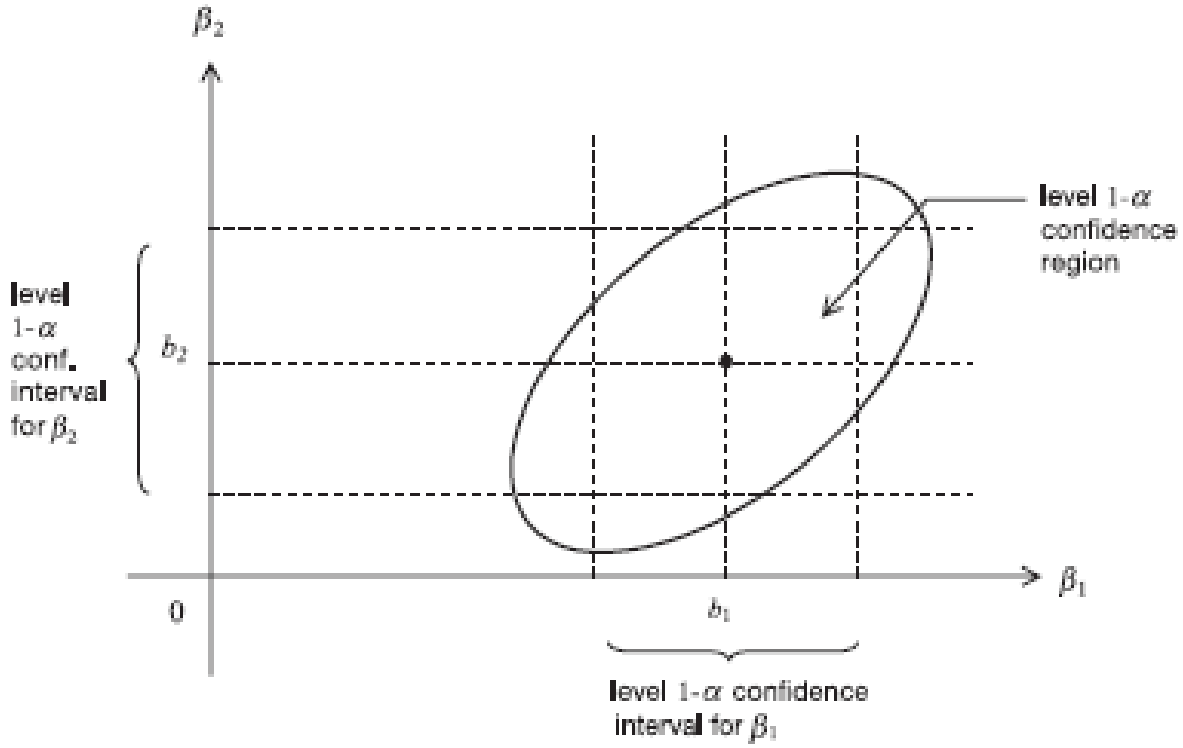

Figure 1.5: t- versus F-Tests

Chow test

- Chow test is a test of whether the true coefficients in two linear regressions on different data sets are equal.
- In econometrics, it is most commonly used in time series analysis to test for the presence of a structural break at a period which can be assumed to be known a priori (for instance, a major historical event such as a war)
- The Chow test tells you if the regression coefficients are different for split data sets.

Chow test example

The equation of interest is

$$sleep_i = \beta_1 + \beta_2 totwrk_i + \beta_3 educ_i + \beta_4 age_i + \beta_5 age_i^2 + \beta_6 yngkid_i + \epsilon_i$$

Test for equality of the parameters in the sleep equation for men and women. Testing procedure:

- $H_0 = \beta_i^m = \beta_i^f, i = 1, ..., 6; H_1 : H_0$ not true

Chow test example

- Estimate the model over the two sub-samples (men and women)
- Compute the following test statistic:

$$F = \frac{(SSR_R - SSR_U)/\#r}{SSR_U/(n-K)} = \frac{(SSR_R - SSR_U)/\#r}{s^2}$$

- In this example:

$$F = \frac{(SSR_R - (SSR_m + SSR_f))/6}{(SSR_m + SSR_f)/(n-K)}$$

- Unrestricted model allows for different parameters for men and women, restricted model assumes the same $\beta's$ so $K = 2k = 2 * 6$, where k denotes the number of parameters in restricted model